



✓ Working with ELK Stack

- ✓ **ELK Stack** contains 3 open-source projects

Elasticsearch,

Logstash

Kibana

- ✓ These 3 open-source projects are implemented by the company called **Elastic**.
- ✓ These three projects are used together for log analysis in various environments

Elasticsearch

- ✓ Elasticsearch is a NoSQL database that is based on the Lucene search engine.
- ✓ Elastic search indexes and store log information

Logstash

- ✓ Logstash is a log pipeline that accepts inputs from various sources,
- ✓ Logstash collects and parses logs

Kibana

- ✓ Kibana is a visualization UI layer that works on top of Elasticsearch.
- ✓ Kibana provides a UI layer that provide actionable insights.

Setup Your Boot Projects:

- 1) Take the WorkSpace : **TopicDemos\ELK-Demo**
- 2) Copy the following 3 apps from **Opt6-Using-Feign-Eureka**
 - a. MyEurekaServer
 - b. BookPriceMS
 - c. BookSearchMS
- 3) Make sure that you are using Logging Statements.

- 4) Add the following Property in application.properties of **BookPriceMS**

logging.file.name=E:\\mylogs\\jlc-book-price.log

- 5) Add the following Property in application.properties of **BookSearchMS**

logging.file.name=E:/mylogs/jlc-book-search.log

Setup ElasticSearch:

- 1) Open **<https://www.elastic.co/downloads/elasticsearch>**
- 2) Download ElasticSearch for Windows
- 3) You will get **elasticsearch-7.15.2-windows-x86_64.zip** file
- 4) Extract the Zip and You will get folder called **elasticsearch-7.15.2**
- 5) Copy **elasticsearch-7.15.2** to E Drive.

Now E:\\elasticsearch-7.15.2 is the Home Directory.

- 6) Goto E:\\elasticsearch-7.15.2\\bin and run the command to Start the ElasticSearch

E:\\elasticsearch-7.15.2/bin> elasticsearch.bat

- 7) Open the following URL in the browser

<http://localhost:9200>

Setup Kibana:

- 1) Open **<https://www.elastic.co/downloads/kibana>**
- 2) Download Kibana for Windows
- 3) You will get **kibana-7.15.2-windows-x86_64.zip** file
- 4) Extract the Zip and You will get folder called **kibana-7.15.2**
- 5) Copy **kibana-7.15.2** to E Drive.

Now E:\\ kibana-7.15.2 is the Home Directory.



6) Update E:\kibana-7.15.2\config\kibana.yml

elasticsearch.hosts: ["http://localhost:9200"]

7) Goto E:\kibana-7.15.2\bin and run the command to Start the Kibana

E:\kibana-7.15.2\bin> kibana.bat

Setup LogStash:

- 1) Open **<https://www.elastic.co/downloads/logstash>**
- 2) Download **logstash** for Windows
- 3) You will get **logstash-7.15.2-windows-x86_64.zip** file
- 4) Extract the Zip and You will get folder called **logstash-7.15.2**
- 5) Copy **logstash-7.15.2** to E Drive.

Now E:\logstash-7.15.2 is the Home Directory.

6) Define **LogStash pipeline** in logStash Configuration file - **jlc-logstash.conf**

7) Goto **E:\logstash-7.15.2\bin** and run the following

E:\logstash-7.15.2\bin> logstash -f jlc-logstash.conf

1. jlc-logstash.conf

```
input {  
  file {  
    type => "java"  
    path => "E:/mylogs/jlc-book-search.log"  
    codec => multiline {  
      pattern => "^%{YEAR}-%{MONTHNUM}-%{MONTHDAY} %{TIME}.*"  
      negate => "true"  
      what => "previous"  
    }  
  }  
}
```



```
}  
}  
file {  
  type => "java"  
  path => "E:/mylogs/jlc-book-price.log"  
  codec => multiline {  
    pattern => "^%{YEAR}-%{MONTHNUM}-%{MONTHDAY} %{TIME}.*"  
    negate => "true"  
    what => "previous"  
  }  
}  
}  
  
filter {  
  if [message] =~ "\tat" {  
    grok {  
      match => ["message", "^(\\tat)"]  
      add_tag => ["my-jlc-error-trace"]  
    }  
  }  
}  
  
output {  
  stdout {  
    codec => rubydebug  
  }  
  
  elasticsearch {  
    hosts => ["localhost:9200"]  
  }  
}
```



Now All 3 ELK Components are running.

- 1) Start **Eureka Server** (Port : 8761)
- 2) Start 1 Instance of **BookPriceMS** (Port : 9000)
- 3) Start 1 Instance of **BookSearchMS** (Port : 8000)
- 4) Open the Browser and hit the following URL
<http://localhost:8000/mybook/101>
<http://localhost:8000/mybook/102>
<http://localhost:8000/mybook/103>

<http://localhost:8000/mybooks/srinivas>
<http://localhost:8000/mybooks/dandes>
- 5) Open the Kibana Workbench
<http://localhost:5601>
- 6) Create Index Patterns
 - a. Select Stack management => Index Patterns
 - b. Click on Create **Index Pattern**
 - c. Provide the following
 - i. **Name : logstash-***
 - ii. **Timestamp field**

Click on Create Index Pattern
- 7) Go to Discover Option in Kibana Workbench
- 8) See the Log Traces.



BookPriceMS – Files Required

1. BookPriceApplication.java	Same
2. BookPriceController.java	Same
3. BookPriceInfo.java	Same
4. application.properties	Updated
5. pom.xml	Same

4. application.properties

spring.application.name=**MyBookPriceMS**

server.port=**9000**

eureka.client.service-url.defaultZone=http://localhost:8761/eureka

eureka.client.healthcheck.enabled=true

logging.level.root=INFO

logging.pattern.console=%-5level %logger{36} - %msg%n

logging.file.name=e:/mylogs/jlc-book-price.log

BookSearchMS – Files Required

1. BookSearchApplication.java	Same
2. BookSearchController.java	Updated
3. BookPriceProxy.java	Same
4. BookPriceInfo.java	Same
5. BookInfo.java	Same
6. application.properties	Updated
7. pom.xml	Same

2. BookSearchController.java

package com.jlcindia.booksearch;

import java.io.PrintWriter;

import java.io.StringWriter;

import org.slf4j.Logger;



```
import org.slf4j.LoggerFactory;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.CrossOrigin;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

@CrossOrigin
@RestController
public class BookSearchController {

    private static final Logger mylogger =
    LoggerFactory.getLogger(BookSearchController.class.getName());

    @GetMapping("/mybook/{bookId}")
    public BookInfo getBookById(@PathVariable Integer bookId) {
mylogger.info("---BookSearchController---getBookById()-----");

        BookInfo bookInfo=new BookInfo();

        bookInfo.setBookId(bookId);
        bookInfo.setBookName("Master Spring MicroServices");
        bookInfo.setAuthor("Srinivas Dande");
        bookInfo.setCategory("Java");
        bookInfo.setPublications("JLC");

        // Need to Invoke BookPriceMS

        BookPriceInfo bookPriceInfo = bookPriceProxy.getBookPrice(bookId);
        bookInfo.setPrice(bookPriceInfo.getPrice());
        bookInfo.setOffer(bookPriceInfo.getOffer());
        bookInfo.setBookPriceServerPort(bookPriceInfo.getBookPriceServerPort());
        bookInfo.setBookSearchServerPort(bookSearchServerPort);

        mylogger.info(bookInfo.toString());
        return bookInfo;
    }
}
```



```
@GetMapping("/mybooks/{author}")
public BookInfo getBookByAuthor(@PathVariable String author) {
mylogger.info("---BookSearchController---getBookByAuthor()-----");
BookInfo bookInfo=new BookInfo();

try {
throw new Exception("Exception in fetching books with Author Name....");
} catch (Exception exception) {
exception.printStackTrace();
LOG.error(exception.toString());

StringWriter sw = new StringWriter();
PrintWriter pw = new PrintWriter(sw);
exception.printStackTrace(pw);
String stackTrace = sw.toString();
mylogger.error("MyException - " + stackTrace);
}

mylogger.info(bookInfo.toString());
return bookInfo;
}
}
```

6. application.properties

```
spring.application.name=MyBookSearchMS

server.port=8000

eureka.client.service-url.defaultZone=http://localhost:8761/eureka
eureka.client.healthcheck.enabled=true

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

logging.file.name=e:/mylogs/jlc-book-search.log
```